**1** **(a)** $x^2 - 5$ **[B1]**

**(b)** $(x-3)^2$ **[B1]**

**(c)** Translation **[B1]**

by vector $\begin{pmatrix} -4 \\ -1 \end{pmatrix}$ **[B1]**

*[Total Marks: 4]*

**2** **(a)** $g(x) = f(x) + 7$ **[M1]**

$g(x) = 2x + 2$ **[A1]**

**(b)** Translation **[B1]**

by vector $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$ **[B1]**

*[Total Marks: 4]*

**3** **(a)** $(-2, 1)$ **[B1]**

**(b)** $(5, -1)$ **[B1]**

**(c)** **(i)** $a = 2$ **[B1]**

$b = -5$ **[B1]**

**(ii)** $(5, -6)$ **[B1]**

(Strict follow through from their vector in (c)(i) applied to $(3, -1)$)

*[Total Marks: 5]*

**4** **(a)** $g(x) = f(x+2) = 3(x+2)^2$ **[M1]**

$3(x^2 + 4x + 4)$ **[M1]**

$3x^2 + 12x + 12$ **[A1]**

**(b)** $(-2, -8)$ **[B2]**

*(Allow B1 for each correct coordinate)*

**(c)** Substitutes $x = 0$ into $h(x) = 3(x+2)^2 - 8$ (or similar valid method) **[M1]**

4 **[A1]**

*[Total Marks: 6]*

**5** **(a)** Finds $(x-3)^2$ as part of completing the square: $f(x) = (x-3)^2 - 9 + 14$ **[M1]**

$f(x) = (x-3)^2 + 5$ **[M1]**

Translation vector $\begin{pmatrix} 3 \\ 5 \end{pmatrix}$ **[A1]**

**(b)** Substitutes $x-4$ into $g(x)$: $g(x-4)=\dfrac{3}{(x-4)+2}-5$ **[M1]**

Adds 2 to the function: $h(x)=\dfrac{3}{x-2}-5+2$ **[M1]**

$h(x)=\dfrac{3}{x-2}-3$ or equivalent single fraction $\dfrac{9-3x}{x-2}$ **[A1]**

**(c)** $x=2$ **[B1]**

$y=-3$ **[B1]**

(Strict follow through from their $h(x)$ in part (b))

*[Total Marks: 8]*

**6 (a) (i)** $\dfrac{20}{x}-4$ **[B1]**

**(ii)** $\dfrac{20}{x}-4=0$ **[M1]**

$x=5$ **[A1]**

**(b)** $h(x)=f(x-k)=\dfrac{20}{x-k}$ **[M1]**

$\dfrac{20}{8-k}=5$ **[M1]**

$k=4$ **[A1]**

**(c)** Vertical asymptote: $x=3$ **[B1]**

Horizontal asymptote: $y=7$ **[B1]**

*[Total Marks: 8]*